Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/25/2022
PI Name:	Gholami, Behnood Ph.D.		
Project Title:	Remote Cardiopulmonary Sensing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07030-4692	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	07/06/2020	**End Date:**	07/05/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC20C0117		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Cardiopulmonary monitoring is of critical importance in a variety of clinical and non-clinical applications ranging from monitoring physiological conditions of crew members during space missions to emotion and stress recognition in applications involving human-machine interaction. Current solutions involve attaching gel-based electrodes for electrocardiogram (ECG) monitoring and pulse oximetry sensors connected to fingertips or earlobes for photoplethysmography (PPG) monitoring. Gel-based electrodes require preparation and their application can cause skin irritation. In addition, the use of current contact-based solutions is further complicated by the fact that a relatively large device such as a Holter monitor has to be carried by the subject at all times. Wearable sensors are a step in the right direction, yet the sensor needs to be continuously worn (on the wrist, chest, etc.) by the subject. We propose to build on our prior research experience in non-invasive remote cardiopulmonary monitoring as well as computer vision and		

	machine learning to develop a non-invasive cardiopulmonary monitoring system and extract clinically important information from multiple subjects in the field of view. Specifically, our proposed sensing framework involves i) an optical camera, ii) a depth-sensing camera, iii) a Doppler radar-based solution, and iv) a sensor fusion component for integration of data received by multiple sensing modalities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2020. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)